Fiscal Year:	FY 2012	**Task Last Updated:**	FY 03/06/2013
PI Name:	Huff, Janice Ph.D.		
Project Title:	Impact of Radiation Quality on Cancer Processes in 2D and 3D Esophageal Cell Models		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
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Space Biology Element:	None		
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Space Biology Special Category:	None		
PI Email:	janice.l.huff@nasa.gov	**Fax:**	FY
PI Organization Type:	INDUSTRY	**Phone:**	281-483-9883
Organization Name:	Wyle Laboratories, Inc./NASA Johnson Space Center		
PI Address 1:	Wyle		
PI Address 2:			
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Space Radiobiology NNJ09ZSA001N
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Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
Contact Email:	lisa.c.simonsen@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date is now 9/1/2013 (from 6/30/2012) per PI (Ed., 10/18/2012) NOTE: End date is now 6/30/2012 (from 12/31/2010) per PI (2/11/2011) NOTE: End date changed to 12/31/2010 (from 11/14/2010), per S. Krenek/JSC (8/2010)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Rustgi, Anil (University of Pennsylvania)		
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Task Description:	Esophageal cancer is the 6th leading cause of cancer death worldwide and the 7th leading cause of cancer death in the U.S. Esophageal squamous cell carcinoma (ESCC) is the most common histological subtype of esophageal cancer, and is a rapidly metastasizing malignancy that is associated with poor patient prognosis (with a 5 year survival rate <20%). Esophageal cancer is associated with a wide variety of risk factors including ionizing radiation, as noted by its increased incidence as a secondary cancer after radiotherapy for primary carcinomas of the head and neck and mediastinal regions, as well as the high excess relative risk for ESCC observed in atomic bomb survivors. Because of the strong link between radiation exposure and the development of esophageal cancer and the high mortality associated with this type of malignancy, it is essential for cancer risk assessment in astronauts to understand the molecular mechanisms of radiation-enhanced carcinogenesis in the esophagus and to understand how radiation quality affects these mechanisms. The objective of this project is to investigate the effects of radiation on the development of esophageal cancer and specifically its influence on intercellular communication in the tissue microenvironment. Cues from the microenvironment can play a pivotal role in determining the fate of precancerous and transformed cells, mediating both tumor-suppressive and tumor-promoting effects through modulation of cell proliferation, cell death, migration, invasion, and recruitment of new blood vessels. We hypothesize that one path whereby radiation may impact esophageal carcinogenesis is through non-targeted mechanisms involving alterations in the stromal fibroblasts that drive preneoplastic changes in esophageal epithelial cells. Our model system consists of normal esophageal fibroblasts and immortalized esophageal epithelial cells and genetic variants grown using standard 2-D culture techniques. We also use a 3-D organotypic model that was developed in the laboratory of project Co-I Dr. Anil Rustgi at the University of Pennsylvania. In this tissue engineering model, esophageal epithelial cells are grown in the presence of fibroblasts in a realistic tissue-like environment that mimics the morphological features, differentiation markers, and growth characteristics observed in normal human tissue. In preliminary experiments, we observed an increase in epithelial cell migration and invasion, both of which are hallmarks of cancer, when esophageal epithelial cells were cultured in the presence of conditioned media from irradiated stromal fibroblasts. To characterize this observation and to establish methodologies to utilize 3-D organotypic culture techniques for radiobiology experimentation, we pursued the following subset of specific aims during the first year of this project: Aim 1 - Determine how radiation alters stromal fibroblast interaction with esophageal epithelial cells by measuring senescence, genome instability, and secretion of bioactive molecules. Aim 2 - Examine how radiation impacts hallmarks of carcinogenesis, specifically genomic instability, in 3-D organotypic cultures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understanding the early molecular mechanisms of carcinogenesis in the esophagus is important not only for risk assessment for astronauts who will encounter excessive amounts of radiation during long duration space travel, but also for applications here on Earth; these findings may contribute to advances in early disease detection and targeted treatment strategies for this rapidly fatal malignancy. In addition, contributions in the area of tissue engineering that result from the use of organotypic cell culture models may translate to and promote progress in other areas of biomedical research.
	Esophageal cancer is the 6th leading cause of cancer death worldwide and the 7th leading cause of cancer death in the U.S. Esophageal squamous cell carcinoma (ESCC) is the most common histological subtype of esophageal cancer, and is a rapidly metastasizing malignancy that is associated with poor patient prognosis (with a 5 year survival rate <20%). Esophageal cancer is associated with a wide variety of risk factors including ionizing radiation, as noted by its increased incidence as a secondary cancer after radiotherapy for primary carcinomas of the head and neck and mediastinal regions, as well as the high excess relative risk for ESCC observed in atomic bomb survivors. Because of this strong link between radiation exposure and the development of esophageal cancer, as well as the high mortality associated with this type of malignancy, it is essential for cancer risk assessment in astronauts to understand the molecular mechanisms of radiation-enhanced carcinogenesis in the esophagus and the impact of radiation quality on these mechanisms. The objective of this project is to investigate the effects of radiation on the development of esophageal cancer and, specifically, the influence of radiation on intercellular communication in the tissue microenvironment. Cues from the microenvironment can play a pivotal role in determining the fate of precancerous and transformed cells, mediating both tumor-suppressive and tumor-promoting effects through modulation of cell proliferation, cell death, migration, invasion, and recruitment of new blood vessels. We hypothesize that one path whereby radiation may impact esophageal carcinogenesis is through non-targeted mechanisms involving alterations in the stromal fibroblasts that drive preneoplastic changes in esophageal epithelial cells. Our model system consists of normal esophageal fibroblasts and immortalized esophageal epithelial cells and genetic variants grown using standard 2-D culture techniques. We also utilize a 3-D organotypic model that was developed in the laboratory of project Co-I Dr. Anil Rustgi at the University of Pennsylvania. In this tissue engineering model, esophageal epithelial cells are grown in the presence of fibroblasts in a realistic, tissue-like environment that mimics the morphological features, differentiation markers, and growth characteristics observed in normal human tissue. In preliminary experiments, we observed an increase in epithelial cell migration and invasion, both of which are hallmarks of cancer, when esophageal epithelial cells were cultured in the presence of conditioned media from irradiated stromal fibroblasts. Antibody-capture cytokine arrays (Raybiotech, Inc.) were used to examine bioactive compounds in the conditioned media of irradiated stromal fibroblasts and allowed identification of hepatocyte growth factor (HGF) as an abundant protein that exhibited increased levels following irradiation with 2 Gy gamma-rays. HGF is a paracrine effector that works in concert with its cognate receptor MET, a transmembrane protein tyrosine kinase that mediates cell- and tissue specific effects on a wide variety of cellular functions including cell proliferation, motility, invasion, cell survival as well as apoptosis. Using ELISAs, we quantified a radiation dose-dependent increase in the levels of HGF in conditioned media of irradiated fibroblasts, and RT-qPCR was used to confirm increased mRNA levels for HGF at 1 hour post-radiation exposure. Recombinant HGF alone caused an increase in the migratory capacity of normal epithelial cells (EPC2-hTERT) in modified Boyden chamber assays and in scratch wound healing assays. Neutralizing antibodies against HGF blocked the enhanced migration and invasion of EPC2-hTERT observed when these cells were co-cultured with irradiated fibroblasts. These results led to the conclusion that HGF was a major mediator of the increased migratory and invasive

Task Progress:	behaviors of these cells observed in our preliminary work. The results were published (Patel, Z.S., Grugan K.D., Rustgi, A.K., Cucinotta, F.A, Huff, J.L. Ionizing radiation enhances esophageal epithelial cell migration and invasion through a paracrine mechanism involving stromal-derived hepatocyte growth factor. Rad. Res. 2012). Increased levels of HGF were also observed in conditioned media from fibroblasts irradiated with high-LET heavy ions (60 rad doses) of different qualities (28Si 250 MeV/amu; 48Ti 350 MeV/amu and 56Fe 600 MeV/amu), indicating that induction of HGF secretion following ionizing radiation exposure was not a low-LET radiation dependent phenomenon. Another important avenue by which the HGF/Met signaling axis modifies cellular behavior is through strong proliferative and pro-survival pathway activation. These actions occur in a cell- and tissue-dependent manner and can have very important implications for radiation carcinogenesis as they may serve to promote proliferation and survival of genetically damaged cells. To investigate these critical pathways of HGF/Met function in our model system we used the cytokinesis-block micronucleus assay to measure radiation-induced genotoxic damage in cells cultured in 2-D monolayers with and with-out the addition of recombinant HGF. Increased accumulation of binucleated cells with micronuclei were observed in HGF-treated cell cultures compared to untreated controls. This effect was evident in samples irradiated with both low- and high-LET radiation. Several scenarios may account for this observation, including changes in the kinetics of DNA repair, alterations in ability of the cell's DNA repair machinery to recognize damaged DNA, changes in cell-cycle kinetics, aberrant checkpoint activation, alterations in mitotic control and/or changes in signaling related to cell survival and apoptosis. No significant difference in the number of ionizing radiation induced foci, a marker of DNA double-strand breaks, were observed in HGF-treated cells at 6 and 24 hours post irradiation with 1 Gy 137Cs gamma-rays. However, using qRT-PCR, suppressed expression of the gene for the pro-apoptotic mediator BBC3 was observed at 24 hours following irradiation with both 137Cs gamma-rays (200 rad) and 48Ti 350 MeV/u ions (30 rad) in cells pretreated with HGF, which suggests an anti-apoptotic role for HGF under these experimental conditions. Further characterization of the modulatory effects of the HGF/Met signaling axis on radiation responses in this model are outside the scope of this project. However this may be a relevant area of future investigation that will provide information on the multiple impacts of stromal-epithelial communication and the role of the microenvironment in tissue responses to radiation exposure. The final area of investigation during this reporting period was to complete sample collection and analysis of radiation-induced genomic damage in 3-D organotypic cultures. In the previous reporting period we developed methods for growth of these cultures at NSRL as well as modified techniques for use of the cytokinesis-block micronucleus assay to measure genetic damage following radiation exposure. The use of these constructs at NSRL presented a logistic challenge due to the long culture time required, both before and after irradiation; and therefore during our initial experimental runs (NSRL10A and 10C) we tested several scenarios to optimize this process. We determined that the most reliable and highest quality cultures required full preparation on-site at BNL, so during subsequent experimental campaigns (NSRL11C and NSRL12A) all cultures were prepared in full at BNL. To assess radiation quality effects on genotoxic damage in these tissue realistic organotypic cultures and 2-D equivalents we collected samples irradiated with a dose range spanning 3 – 120 rad from heavy ion beams that were selected to cover a range of LETs and energies. We are in the process of data collection and analysis, and expect that this data-set will provide useful information for NASA for the purpose of risk assessment of the effects of high-LET particle radiation, including exposure to heavy ions at low fluence where there is less than one particle track traversal per cell.

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